Computational Logic

Propositional Logic ct’d

Jakub Szymanik

- Week 1 (14.09): Intro (try to read Logicomix after)
- Week 2 (21.09): How to Win a Million Dollars with Logic? Part 1:The language and what it means. (Study Sections 1.1, 1.3, 1.4.1-1.4.2 of the book)
- Week 3: (28.09) How to win a million dollars with logic? Part 2: Proof - and why it matches meaning. (Study 1.2, 1.4.3-1.4.4)
- Week 4: (05.10) How to win a million dollars with logic? Part 3: Normal forms and solvers. Lab, BY laptops (Study 1.5, 1.6)
- Homework 1.
- Week 5: (12.10) FO and non-computability (Study 2.1-2.2)
- Week 6: (19.10) FO and non-computability (Study 2.3-2.4)
- Week 7: (26.10) FO and non-computability (Study 2.5) + Trial intermediate exam.
- Homework 2.
- Week 8: (02.11) Expressiveness, Definability, and Generalized Quantifier Theory (Study 2.6-2.7, and the book chapter on Moodle)
- Week 9: (09.11) Modal Logic and Agency 1 (Study 5.2-5.3)
- Week 10: (16.11) Modal Logic and Agency 2 (Study 5.4-5.5)
- Week 11: (23.11) Modal Logic and Agency 3.
- Homework 3.
- Week 12: (30.11) Q&A + Trial final exam.

How are the exercises 1.1,1.3, and 1.4?

Last week

- We defined PL inductively as a formal language
- We discussed the compositionality principle
- We introduced parse trees
- And basic semantic notions: validity, tautology, etc
- And semantic truth-table-method
- And we were careful about what we had not done: we have no way yet to derive anything. Everything last week was a claim about all 2^n rows. Today we build the other turnstile. Why?

NATURAL DEDUCTION FOR PROPOSITIONAL LOGIC

CONJUNCTION

$$\frac{\varphi \quad \psi}{\varphi \wedge \psi}\ \wedge i \qquad \frac{\varphi \wedge \psi}{\varphi}\ \wedge e_1 \qquad \frac{\varphi \wedge \psi}{\psi}\ \wedge e_2$$

DISJUNCTION

$$\frac{\varphi}{\varphi \vee \psi}\ \vee i_1 \qquad \frac{\psi}{\varphi \vee \psi}\ \vee i_2 \qquad \frac{\varphi \vee \psi \qquad \begin{array}{|c|}\hline \varphi \\ \vdots \\ \chi \\ \hline \end{array} \qquad \begin{array}{|c|}\hline \psi \\ \vdots \\ \chi \\ \hline \end{array}}{\chi}\ \vee e$$

IMPLICATION

$$\frac{\begin{array}{|c|}\hline \varphi \\ \vdots \\ \psi \\ \hline \end{array}}{\varphi \to \psi}\ \to i \qquad \frac{\varphi \quad \varphi \to \psi}{\psi}\ \to e$$

NEGATION AND ⊥

$$\frac{\begin{array}{|c|}\hline \varphi \\ \vdots \\ \bot \\ \hline \end{array}}{\neg \varphi}\ \neg i \qquad \frac{\varphi \quad \neg \varphi}{\bot}\ \neg e \qquad \frac{\bot}{\varphi}\ \bot e \qquad \frac{\neg \neg \varphi}{\varphi}\ \neg \neg e$$

DERIVED

$$\frac{\varphi \to \psi \quad \neg \psi}{\neg \varphi}\ MT \qquad \frac{\varphi}{\neg \neg \varphi}\ \neg \neg i \qquad \frac{\begin{array}{|c|}\hline \neg \varphi \\ \vdots \\ \bot \\ \hline \end{array}}{\varphi}\ PBC \qquad \frac{}{\varphi \vee \neg \varphi}\ LEM$$

copy — any line still in scope may be repeated. A box may be cited only after it closes, and only as its conclusion.

Excercise 1.2

Exercises 1.2, p. 78 — starred items have worked solutions on Moodle

- Tier 1 — $\wedge$ and $\rightarrow$ only, no boxes yet: 1(a), 1(c)
- Tier 2 — your first boxes, $\rightarrow$i: 1(f), 1(h), 1(i)
- Tier 3 — negation, $\bot$, MT: 1(e), 2(g)
- Tier 4 — $\vee$e and nested boxes: 2(f), 2(i), 3(n)
- Tier 5 — strategy, LEM and PBC: 3(q), 5(a), 5(d) [take home]

Stuck? Look at the goal, not the premises. The goal picks the rule.

How to actually find a proof?

- Look at the goal first. The goal picks the rule.
- Goal is $\phi \to \psi$ → open a box, assume ϕ, aim for ψ ($\to$i — almost always the right first move)
- Goal is $\phi \wedge \psi$ → prove each half separately ($\wedge$i)
- Goal is $\neg\phi$ → open a box, assume ϕ, aim for $\bot$ ($\neg$i)
- Goal is $\phi \vee \psi$ → try $\vee$i; if that fails you need $\bot$e, or PBC
- Goal is an atom, or you're stuck → work forwards from the premises ($\wedge$e, $\to$e, MT); failing that, PBC: assume ¬goal, aim for $\bot$
- A premise is a disjunction → $\vee$e. Both branches must end at the same conclusion.
- Work from both ends and meet in the middle. Leave a gap and fill it.

Excercise 1.2

Exercises 1.2, p. 78 — starred items have worked solutions on Moodle

- Tier 1 — $\wedge$ and $\rightarrow$ only, no boxes yet: 1(a), 1(c)
- Tier 2 — your first boxes, $\rightarrow$i: 1(f), 1(h), 1(i)
- Tier 3 — negation, $\bot$, MT: 1(e), 2(g), **in pairs**
- Tier 4 — $\vee$e and nested boxes: 2(f), 2(i), 3(n)
- Tier 5 — strategy, LEM and PBC: 3(q), 5(a), 5(d) [take home]

Stuck? Look at the goal, not the premises. The goal picks the rule.

Five ways to lose marks

- Citing a line inside a closed box — once a box closes, only its conclusion survives
- $\vee$e where the two branches end at different conclusions
- Using $\rightarrow$e backwards: from $\phi \rightarrow \psi$ and ψ, concluding ϕ — that's the server-outage fallacy from last week
- Writing $\models$ when you mean $\vdash$
- An unlabelled line. Every line needs its rule and the lines it came from.

Excercise 1.2

Exercises 1.2, p. 78 — starred items have worked solutions on Moodle

- Tier 1 — $\wedge$ and $\rightarrow$ only, no boxes yet: 1(a), 1(c)
- Tier 2 — your first boxes, $\rightarrow$i: 1(f), 1(h), 1(i)
- Tier 3 — negation, $\bot$, MT: 1(e), 2(g), **in pairs**
- Tier 4 — $\vee$e and nested boxes: 2(f), 2(i), 3(n) - on your own pace
- Tier 5 — strategy, LEM and PBC: 3(q), 5(a), 5(d) [take home]

Stuck? Look at the goal, not the premises. The goal picks the rule.

Theorem 1.35 (Soundness) *Let* $\phi_1, \phi_2, \ldots, \phi_n$ *and* ψ *be propositional logic formulas. If* $\phi_1, \phi_2, \ldots, \phi_n \vdash \psi$ *is valid, then* $\phi_1, \phi_2, \ldots, \phi_n \models \psi$ *holds.*

- Let’s check some of the theorems you have proved…
- $p \wedge q \models p$
- $p \vee q \models p$

Proof

Mathematical induction on the length of the proof

'For all sequents $\phi_1, \phi_2, \ldots, \phi_n \vdash \psi$ $(n \geq 0)$ *which have a proof of length* k*, it is the case that* $\phi_1, \phi_2, \ldots, \phi_n \vDash \psi$ *holds.'*

- On the blackboard
- Pick the possible last rule and prove the inductive step for it

1	$p \wedge q \to r$	premise
2	p	assumption
3	q	assumption
4	$p \wedge q$	$\wedge$i 2, 3
5	r	$\to$e 1, 4
6	$q \to r$	$\to$i 3–5
7	$p \to (q \to r)$	$\to$i 2–6

1	$p \wedge q \to r$	premise
2	p	premise
3	q	assumption
4	$p \wedge q$	$\wedge$i 2, 3
5	r	$\to$e 1, 4
6	$q \to r$	$\to$i 3–5

This is a proof of the sequent $p \wedge q \to r,\ p \vdash p \to r$. The induction hypothesis then ensures that $p \wedge q \to r,\ p \models p \to r$ holds. But then we can also reason that $p \wedge q \to r \models p \to (q \to r)$ holds as well – why?

'For all sequents $\phi_1, \phi_2, \ldots, \phi_n \vdash \psi$ $(n \geq 0)$ *which have a proof of length* k*, it is the case that* $\phi_1, \phi_2, \ldots, \phi_n \models \psi$ *holds.'*

Look up the proof on p. 48, the other cases.

Theorem 1.35 (Soundness) *Let $\phi_1, \phi_2, \ldots, \phi_n$ and ψ be propositional logic formulas. If $\phi_1, \phi_2, \ldots, \phi_n \vdash \psi$ is valid, then $\phi_1, \phi_2, \ldots, \phi_n \models \psi$ holds.*

How do you use it to show that a particular proof doesn't exist?

Theorem 1.37 *If $\vDash \eta$ holds, then $\vdash \eta$ is valid. In other words, if η is a tautology, then η is a theorem.*

Exercises 1.4.17

Corollary 1.39 (Soundness and Completeness) *Let* $\phi_1, \phi_2, \ldots, \phi_n, \psi$ *be formulas of propositional logic. Then* $\phi_1, \phi_2, \ldots, \phi_n \models \psi$ *is holds iff the sequent* $\phi_1, \phi_2, \ldots, \phi_n \vdash \psi$ *is valid.*

What it says, and why it isn't obvious

- Soundness, which we just proved: $\vdash$ implies $\models$ — the rules never lie
- Completeness, now: $\models$ implies $\vdash$ — the rules miss nothing
- Look at how different those two sides are:
- $\models$ is a claim about all 2^n valuations
- $\vdash$ is a claim that one finite object exists
- Nothing we have built so far connects them
- What follows is a procedure that turns the first into the second

The plan — three steps

Want: $\phi_1, \ldots, \phi_n \vDash \psi \Longrightarrow \phi_1, \ldots, \phi_n \vdash \psi$

① Bundle. Put $\eta := \phi_1 \to (\phi_2 \to (\ldots \to (\phi_n \to \psi) \ldots))$. If the sequent holds semantically, η is a tautology.

② The work. Every tautology is provable: $\vDash \eta \Longrightarrow \vdash \eta$

③ Unbundle. From $\vdash \eta$, n applications of $\to$e with $\phi_1, \ldots, \phi_n$ give $\phi_1, \ldots, \phi_n \vdash \psi$

① and ③ are bookkeeping. ② is the theorem.

Step ② — one row of the table, one derivation

- Fix one row of the truth table for η, over atoms $p_1, \ldots, p_n$
- Write down that row as literals: $\hat{p}_i$ is p_i if the row makes p_i true, $\neg p_i$ if false
- Claim. For every formula ϕ and every row:
- row makes ϕ true $\Longrightarrow \hat{p}_1, \ldots, \hat{p}_n \vdash \phi$
- row makes ϕ false $\Longrightarrow \hat{p}_1, \ldots, \hat{p}_n \vdash \neg\phi$
- Proved by structural induction on ϕ — one case per connective
- Each case is a three- or four-line derivation. You could write them yourselves.

The implication case

$\phi = \phi_1 \to \phi_2$, row makes it false. Then ϕ_1 true, ϕ_2 false; induction gives $\vdash \phi_1$ and $\vdash \neg\phi_2$.

- assume $\phi_1 \to \phi_2$ · $\to$e gives ϕ_2 · $\neg$e gives $\bot$ · $\neg$i discharges: $\vdash \neg(\phi_1 \to \phi_2)$

$\phi = \phi_1 \to \phi_2$, row makes it true because ϕ_1 is false. Induction gives $\vdash \neg\phi_1$.

- assume ϕ_1 · $\neg$e gives $\bot$ · $\bot$e gives ϕ_2 · $\to$i discharges: $\vdash \phi_1 \to \phi_2$

Gluing the rows together — one atom per round

- η is a tautology, so every row makes it true — every row gives $\hat{p}_1, \ldots, \hat{p}_n \vdash \eta$
- Take the two rows that differ only in the last literal
- LEM gives $p_n \vee \neg p_n$; $\vee$e runs both as cases and discharges both
- What's left: $\hat{p}_1, \ldots, \hat{p}_{n-1} \vdash \eta$ — one atom shorter
- Repeat. n rounds, one atom each. At the end: $\vdash \eta$, nothing on the left.

The whole thing, on one atom

η = (p → ¬p) → ¬p (check: it's a tautology — two rows)
Row p = 1, literal p

1	p	premise
2	| p → ¬p	assumption
3	| ¬p	→e 2, 1
4	(p → ¬p) → ¬p	→i 2–3

Row p = 0, literal ¬p

1	¬p	premise
2	| p → ¬p	assumption
3	| ¬p	copy 1
4	(p → ¬p) → ¬p	→i 2–3

Glue: LEM gives p ∨ ¬p; ∨e over the two boxes ⟹ ⊢ (p → ¬p) → ¬p

Those proofs are exponential

- The proof this procedure builds has size exponential in the number of atoms
- Completeness says a proof exists. It says nothing about a proof being short.
- Is there a proof system in which every tautology has a short proof?
- That question is exactly: NP = coNP? (Cook & Reckhow, 1979) — still open
- So next week we don't search for proofs. We build solvers.
- Week 1: checking is easy, finding is hard. Third time you've met it.

Excercise 1.2

Exercises 1.2, p. 78 — starred items have worked solutions on Moodle

- Tier 1 — $\wedge$ and $\rightarrow$ only, no boxes yet: 1(a), 1(c)
- Tier 2 — your first boxes, $\rightarrow$i: 1(f), 1(h), 1(i)
- Tier 3 — negation, $\bot$, MT: 1(e), 2(g), **in pairs**
- Tier 4 — $\vee$e and nested boxes: 2(f), 2(i), 3(n)
- Tier 5 — strategy, LEM and PBC: 3(q), 5(a), 5(d) [take home]

Stuck? Look at the goal, not the premises. The goal picks the rule.